Alexander Scales

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github.com/AlecScales • RowELO live demo: alecscales.github.io/rowelo

PROFILE

First-year BSc Computer Science student at the University of Bristol with A-levels in Computer Science, Mathematics, Further Mathematics and History. Built RowELO, an end-to-end scraping, database and Elo-ranking platform for British junior rowing, for a client. Seeking spring week and internship opportunities in software engineering and technology.

EDUCATION

BSc Computer Science | University of Bristol, *2026 – 2029*

- Year 1 modules: Imperative & Functional Programming, Mathematics for Computer Science, Computer Architecture, Software Tools

A-Levels | Haberdashers' Monmouth School, *2019 – 2026*

- Computer Science (A), Mathematics (A), Further Mathematics (A), History (A)
- Winner, USW Cyber Security Challenge • Gold, UK Bebras Computational Thinking Challenge • Gold, UKMT Senior Mathematical Challenge

GCSEs: 10 GCSEs including 6 A*s (Mathematics and English Language), 2 As and 2 Bs

FEATURED PROJECT

RowELO: Automated Data Processing & Elo Ranking System | *A-Level NEA, 66/75 (88%)*

- Built a full pipeline that scrapes, cleans, stores and ranks British junior rowing results: 228 races across 18 events and 78 clubs from three regattas over six seasons (2020/21–2025/26); designed to extend to future seasons.
- Designed a 3NF MySQL database covering both multi-lane and head-to-head races.
- Implemented a pairwise Elo algorithm to produce dynamic club rankings, and wrote asynchronous scrapers with data-cleaning logic to normalise club names and inconsistent result formats.
- Generated a static site with a league table, club pages and interactive Elo progression charts (Matplotlib, mpld3).
- Developed with the Head of Rowing at Monmouth School Rowing Club, who confirmed it met their needs and said they would use it regularly to benchmark crews and prepare for races.

EXPERIENCE

Freelance Web Developer | Local organisations

- Gathered requirements and designed and delivered three websites: Monmouth Castle Museum, Made at the Hut and English Bicknor Local History Group.

Hardware Diagnostician (Volunteer) | Repair Café Wales

- Methodically isolated faults, diagnosed components and carried out physical repairs on computers and electronics.
- Explained technical problems and repair options clearly to non-technical members of the public.

Engineering Work Experience | MBDA

- Part of a team of four that won best group project for a design for neutralising an enemy satellite while creating zero space debris.
- Attended sessions on system design, cybersecurity and innovative thinking.

TECHNICAL SKILLS

Languages: Python, C, Haskell, SQL, HTML/CSS
Tools & libraries: MySQL, Git/GitHub, Linux, asynchronous web scraping (requests_html, BeautifulSoup), Matplotlib, mpld3